S001-PROPOSITION-EFFECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter etale.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \external-document list are omitted because the assembler supplies the class.

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Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-proposition-effective.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original equation 0BTI: sources/etale.tex, source lines 2035–2035. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0BTG: sources/algebra.tex, source lines 32506–32506. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 01ZM: sources/limits.tex, source lines 2370–2370. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 07RP: sources/limits.tex, source lines 1949–1949. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 052A: sources/morphisms.tex, source lines 5172–5172. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 02W8: sources/more-morphisms.tex, source lines 16279–16279. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02KQ: sources/descent.tex, source lines 5471–5471. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02KU: sources/descent.tex, source lines 5581–5581. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02NW: sources/morphisms.tex, source lines 13192–13192. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 0AGU: sources/more-algebra.tex, source lines 3087–3087. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 09ZL: sources/more-algebra.tex, source lines 3383–3383. Navigation pointer used in the counted span; the span contains the

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Source reference. Original reference 09XK: sources/more-algebra.tex, source lines 2753–2753. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BTN: sources/etale.tex, source lines 2200–2200. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 01JA: sources/schemes.tex, source lines 2510–2510. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $f : X \to S$ be a surjective integral morphism. The functor (0BTI) induces an equivalence*

$$\begin{array}{c}\textit{schemes quasi-compact,}\\ \textit{separated, étale over } S\end{array} \longrightarrow \begin{array}{c}\textit{descent data } (V,\varphi) \textit{ relative to } X/S \textit{ with}\\ V \textit{ quasi-compact, separated, étale over } X\end{array}$$

Proof. By Lemma 0BTL the functor (0BTI) is fully faithful and the same remains the case after any base change $S \to S'$. Let (V, φ) be a descent data relative to X/S with $V \to X$ quasi-compact, separated, and étale. We can use Lemma 0BTM to see that it suffices to prove the effectivity Zariski locally on S. In particular we may and do assume that S is affine.

If S is affine we can find a directed set Λ and an inverse system $X_\lambda \to S_\lambda$ of finite morphisms of affine schemes of finite type over $\mathrm{Spec}(\mathbf{Z})$ such that $(X \to S) = \lim(X_\lambda \to S_\lambda)$. See Algebra, Lemma 0BTG. Since limits commute with limits we deduce that $X \times_S X = \lim X_\lambda \times_{S_\lambda} X_\lambda$ and $X \times_S X \times_S X = \lim X_\lambda \times_{S_\lambda} X_\lambda \times_{S_\lambda} X_\lambda$. Observe that $V \to X$ is a morphism of finite presentation. Using Limits, Lemmas 01ZM we can find an λ and a descent datum $(V_\lambda, \varphi_\lambda)$ relative to X_λ/S_λ whose pullback to X/S is (V, φ). Of course it is enough to show that $(V_\lambda, \varphi_\lambda)$ is effective. Note that V_λ is quasi-compact by construction. After possibly increasing λ we may assume that $V_\lambda \to X_\lambda$ is separated and étale, see Limits, Lemma 01ZQ and 07RP. Thus we may assume that f is finite surjective and S affine of finite type over $\mathbf{Z}$.

Consider an open $S' \subset S$ such that the pullback (V', φ') of (V, φ) to $X' = X \times_S S'$ is effective. Below we will prove, that $S' \neq S$ implies there is a strictly larger open over which the descent datum is effective. Since S is Noetherian (and hence has a Noetherian underlying topological space) this will finish the proof. Let $\xi \in S$ be a generic point of an irreducible component of the closed subset $Z = S \setminus S'$. If $\xi \in S'' \subset S$ is an open over which the descent datum is effective, then the descent datum is effective over $S' \cup S''$ by the glueing argument of the first paragraph. Thus in the rest of the proof we may replace S by an affine open neighbourhood of ξ.

After a first such replacement we may assume that Z is irreducible with generic point Z. Let us endow Z with the reduced induced closed subscheme structure. After another shrinking we may assume $X_Z = X \times_S Z = f^{-1}(Z) \to Z$ is flat, see Morphisms, Proposition 052A. Let (V_Z, φ_Z) be the pullback of the descent datum

to X_Z/Z. By More on Morphisms, Lemma 02W8 this descent datum is effective and we obtain an étale morphism $U_Z \to Z$ whose base change is isomorphic to V_Z in a manner compatible with descent data. Of course $U_Z \to Z$ is quasi-compact and separated (Descent, Lemmas 02KQ and 02KU). Thus after shrinking once more we may assume that $U_Z \to Z$ is finite, see Morphisms, Lemma 02NW.

Let $S = \mathrm{Spec}(A)$ and let $I \subset A$ be the prime ideal corresponding to $Z \subset S$. Let (A^h, IA^h) be the henselization of the pair (A, I). Denote $S^h = \mathrm{Spec}(A^h)$ and $Z^h = V(IA^h) \cong Z$. We claim that it suffices to show effectivity after base change to S^h. Namely, $\{S^h \to S, S' \to S\}$ is an fpqc covering ($A \to A^h$ is flat by More on Algebra, Lemma 0AGU) and by More on Morphisms, Lemma 02W8 we have fpqc descent for separated étale morphisms. Namely, if $U^h \to S^h$ and $U' \to S'$ are the objects corresponding to the pullbacks (V^h, φ^h) and (V', φ'), then the required isomorphisms

$$U^h \times_S S^h \to S^h \times_S V^h \quad \text{and} \quad U^h \times_S S' \to S^h \times_S U'$$

are obtained by the fully faithfulness pointed out in the first paragraph. In this way we reduce to the situation described in the next paragraph.

Here $S = \mathrm{Spec}(A)$, $Z = V(I)$, $S' = S \setminus Z$ where (A, I) is a henselian pair, we have $U' \to S'$ corresponding to the descent datum (V', φ') and we have a finite étale morphism $U_Z \to Z$ corresponding to the descent datum (V_Z, φ_Z). We no longer have that A is of finite type over $\mathbf{Z}$; but the rest of the argument will not even use that A is Noetherian. By More on Algebra, Lemma 09ZL we can find a finite étale morphism $U_{fin} \to S$ whose restriction to Z is isomorphic to $U_Z \to Z$. Write $X = \mathrm{Spec}(B)$ and $Y = V(IB)$. Since (B, IB) is a henselian pair (More on Algebra, Lemma 09XK) and since the restriction $V \to X$ to Y is finite (as base change of $U_Z \to Z$) we see that there is a canonical disjoint union decomposition

$$V = V_{fin} \amalg V_{away}$$

were $V_{fin} \to X$ is finite and where V_{away} has no points lying over Y. See Lemma 0BTN. Using the uniqueness of this decomposition over $X \times_S X$ we see that φ preserves it and we obtain

$$(V, \varphi) = (V_{fin}, \varphi_{fin}) \amalg (V_{away}, \varphi_{away})$$

in the category of descent data. By More on Algebra, Lemma 09ZL there is a unique isomorphism

$$X \times_S U_{fin} \longrightarrow V_{fin}$$

compatible with the given isomorphism $Y \times_Z U_Z \to V \times_X Y$ over Y. By the uniqueness we see that this isomorphism is compatible with descent data, i.e., $(X \times_S U_{fin}, can) \cong (V_{fin}, \varphi_{fin})$. Denote $U'_{fin} = U_{fin} \times_S S'$. By fully faithfulness we obtain a morphism $U'_{fin} \to U'$ which is the inclusion of an open (and closed) subscheme. Then we set $U = U_{fin} \amalg_{U'_{fin}} U'$ (glueing of schemes as in Schemes, Section 01JA). The morphisms $X \times_S U_{fin} \to V$ and $X \times_S U' \to V$ glue to a morphism $X \times_S U \to V$ which is the desired isomorphism. $\square$